Kidus Workneh Abebe

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Education

St Mary's University, Addis Ababa **June 2021 – Feb 2025**

Bachelor of Science, Computer Science, 3.77 GPA
Computer programming, Database Systems, Data Structures and Algorithms,
Web Development. Operating systems and Computer Architecture.

Professional Experience

Shega Media and Technology, Addis Ababa **August 2025 – July 2026**

Software Engineer

- Architected and deployed highly performant backend APIs using **FastAPI** and **NestJS**, optimizing server-side logic for scalable digital platforms.
- Engineered production-grade AI-driven backend systems, implementing **RAG pipelines and autonomous agents** utilizing **LangChain/LangGraph**, LLMs, and vector databases to streamline high-throughput data retrieval.
- Designed complex relational database schemas and managed data persistence layers utilizing **PostgreSQL** and **Prisma ORM**, enforcing strict transaction atomicity and data consistency.
- Containerized application environments using **Docker** and managed deployment pipelines, leveraging **Linux server administration** and bash scripting for secure, automated remote server management.
- Contributed to specialized machine learning workflows, handling large-scale data preprocessing, feature engineering, and model experimentation.

Technical Projects

Freelancing Website

- Designed the entire backend infrastructure and relational database schema for a multi-user marketplace platform using **nodejs, react** and **mongodb**.
- Implemented secure user authentication, role-based access control, and highly optimized search queries for service listings.

PDF RAG Application

- Developed an advanced distributed RAG backend enabling multi-document processing and contextual retrieval.
- Implemented a multi-tier database strategy featuring **MongoDB** for document metadata storage and a high-performance vector database (**ChromaDB**) for semantic search embeddings

Chat Application

- Built a highly scalable, event-driven real-time messaging server using **Socket.io** and **Node.js**.
- Integrated **Redis** to manage pub/sub-architectures, handle bidirectional communication streams, and maintain stateless user session caching.

Skills

Programming languages: Python, C++, Typescript, Javascript, SQL, Bash Script.

Backend & Frameworks: FastAPI, NestJS, Node.js, Express, Django, Next.js (Backend).

Databases & Caching: PostgreSQL, Prisma ORM, Redis, MongoDB, ChromaDB.

DevOps & Infrastructure: Docker, Linux Server Administration, Git/GitHub, CI/CD, Shell Scripting.

Machine Learning: LangChain, LangGraph, Scikit-learn, TensorFlow, PyTorch.

Languages: Amharic(native), English(Fluent)

Certifications

Supervised Machine Learning And Regression | 🔗 **Certificate**

Neural Networks and Deep Learning | 🔗 **Certificate**